

We can get a better understanding of these boards by going through their detailed specifications given in Table I.

Recently launched in India by ST-Microelectronics, the Nucleo development board features the STM32 MCU family line and allows one to choose the right MCU from amongst the various combinations of power consumption, performance and features. It supports the Arduino connectivity ST Morpho headers that make it easy to expand the functionality of the development platform for Nucleo to choose any specialised shielding. It has an integrated programmer or debugger, and therefore does not need any separate probe for the same. It comes with the STM32 comprehensive HAL library of software and other software examples, along with some online resources.

Tiny development boards

The requirement of compact devices with maximum efficiency and power led to the development of several tiny, Lilliputian-sized development boards. Despite their size, the boards have wireless communication facilities built-in, encouraging the connection of embedded devices and revolutionising the term Internet of Things (IoT). Some of these are described below.

Femtoduino (price: \$22). Considered to be the clone of Arduino, Femtoduino is ultra-small (20.7x15.2 mm) and uses Arduino libraries and compiler. It has a computing power similar to Arduino UNO and uses ATMEGA 328p, 0.13-centimetre connectors, 0402 components and weighs only about two grams. It has 0.25-centimetre-sized pins, and thus can be used on a standard breadboard or perfboard.

Intel Edison. Compatible with open source software tools, Edison features Intel's Atom, a 22nm Silvermont micro-architecture based SoC. It has a dual-core CPU and a single-core MCU with integrated Wi-Fi, Bluetooth low energy (BLE) memory and storage. Edison supports more than 30 industry-standard I/O interfaces via a 70-pin connector. It also supports projects like Arduino, Python, Node.js, Yocto Linux

Basic components of a development board

1. Microcontroller or microprocessor for suitable type of applications
2. Power circuit for supplying power to the MCU and other on-board components
3. Basic inputs, such as buttons, sensors, etc
4. Basic outputs, such as LEDs, LCDs, motors, etc
5. Programming interface to facilitate programming of the MCU
6. Communication and user interface, such as the USB port, wireless network ports, etc

and Worlfram.

Digispark (price: \$8.95). A USB development board with Attiny85 microcontroller, it is also similar to the Arduino board, yet is smaller and cheaper. It supports Arduino IDE 1.0 and is compatible with Linux, Windows and MAC operating systems. It has six I/O pins, of which two are reserved for USB only—this is in case the program communicates over USB. It automatically selects the required power from the external source connected via USB.

mbed LPC1114 (price: \$44.95). A low-cost development board with the NXP LPC1114 MCU, it has been mainly designed for rapid prototyping of low-cost USB devices. But it can also be used for battery-powered applications and 32-bit ARM Cortex-M0-based designs. It has a built-in drag-and-drop Flash programmer with 32kB Flash. It uses a proprietary online compiler that is given access to when the board is purchased. Although this prohibits the need to download and install anything, it encourages the requirement of an active Internet connection to push the codes to the board.

FLORA (price: \$24.95). Designed to empower wearable projects, FLORA is round in shape and has a diameter of 4.45 centimetres. It has a flexible and easy-to-use power supply with a 2A power FET, which is usually not included in wearable options. FLORA is fully Arduino compatible, and so, all the Arduino tutorials and libraries are supposed to work on it.

Teensy 3.1 (price: \$19.80). An ARM Cortex M4-based development board at such low price is a steal. 18mm x 30.5mm in size, Teensy 3.1 is a complete USB-based MCU board. It allows single push-button programming and works with MAC, Linux and Windows operating systems. It supports a Wi-

nAVR C compiler and Teensyduino—an add-on for Arduino IDE. It has about 34 I/O pins suitable for a solder-less breadboard.

USB Bit Whacker (price: \$24.95). The USB Bit Whacker is a PIC-processor-based development board featuring the PIC18F2553 MCU. It is 25mm x 40mm in size and has 16 I/O ports. It produces HEX files with almost any compiler. With the transmission and reception ports, it can be used as a USB to UART converter board.

Mini-32 (price: \$29). Another PIC-based development board, it features the PIC32MX534F064H MCU and operates on 3.3V power supply. It supports CAN communication and can be used as a DIP40 PIC32 substitute for PIC18 and PIC16 MCUs. It has a USB HID boot-loader, which simply means that no additional software is needed to access the board. The compiled files can be simply dragged and dropped.

Other development boards

There are some development boards that do not allow any modification in the existing design and the codes, and they are sometimes compatible with only a few specific software. These boards have very specific development tools that are available either at a very high price or only through specific manufacturers and distributors. These development resources also have very limited support systems.

Most of these boards are available as packaged evaluation kits that contain the development boards along with the required tools and software. Some of these are:

Tiva C series. It features the Tiva TM4C129x-series ARM Cortex-M4-based MCU. It supports the evaluation version of Keil RealView MCU development kit along with IAR Embedded